

CSE350: Digital Electronics & Pulse Techniques

Lecture 07: DTL Power, HTL, TTL Intro

Ex-1: Calculate the maximum power dissipation in a DTL NAND gate.

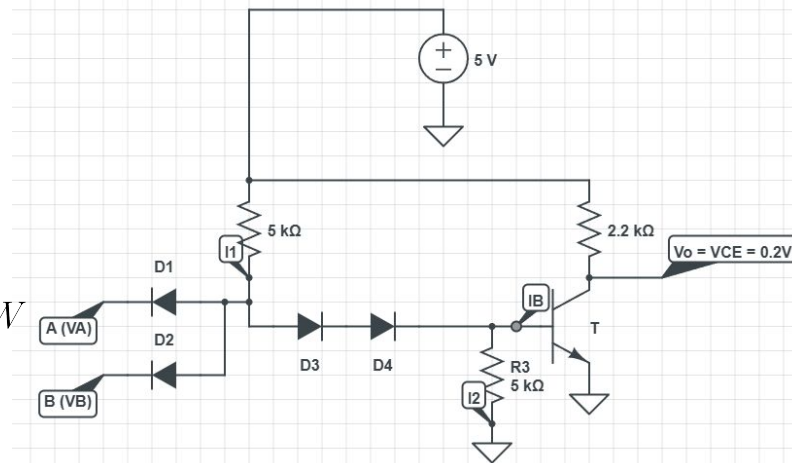
Case(1,1)(Output low): The states of different components & values of different currents, voltages as calculated in previous lectures are summarized below:

Component/ Parameter	D1	D2	D3	D4	T	V_P	I_1	I_C	V_o
State	Off	Off	On	On	Saturation	2.2V	0.56mA	2.182mA	0.2V

So, the power is being supplied by the 5V source, from which two currents are drawn to the circuit and eventually to ground.

Since, Power supplied = Power consumed/dissipated,

Power dissipated in this case $= (5 - 0) * (0.56 + 2.182) = 13.71mW$



Case(0,1|1,0|0,0)(Output low): A similar table as in the previous case would be:

Component/ Parameter	D1	D2	D3	D4	T	V_P	I_1	I_C	V_o
State	On	On	Off	Off	Cutoff	0.9V	0.82mA	0mA	5V

Power dissipation in all three possible input cases would be the same(because T is off in all cases). So, we have considered case(0,0) here. Again we use the concept that, Power supplied = Power consumed/dissipated. Here, 5V is supplying 0.82mA current to the input diodes, which gets split equally at 'P'. Then the currents flow in both branches up to 0.2V of the inputs.

$$\text{So, Power dissipated} = (5 - 0.2) * 0.82 = 3.936mW$$

HTL(High Threshold Circuit): A modified DTL circuit to increase noise margin.

Ex-2: If all inputs are high, then what is the magnitude of Noise Voltage that would cause the HTL circuit to malfunction?

The HTL circuit operates at high voltage. It has a transistor instead of diode like the modified DTL circuit. However, it also doesn't have a second diode. Instead a zener diode is used.

As like the previous example on DTL noise margin,

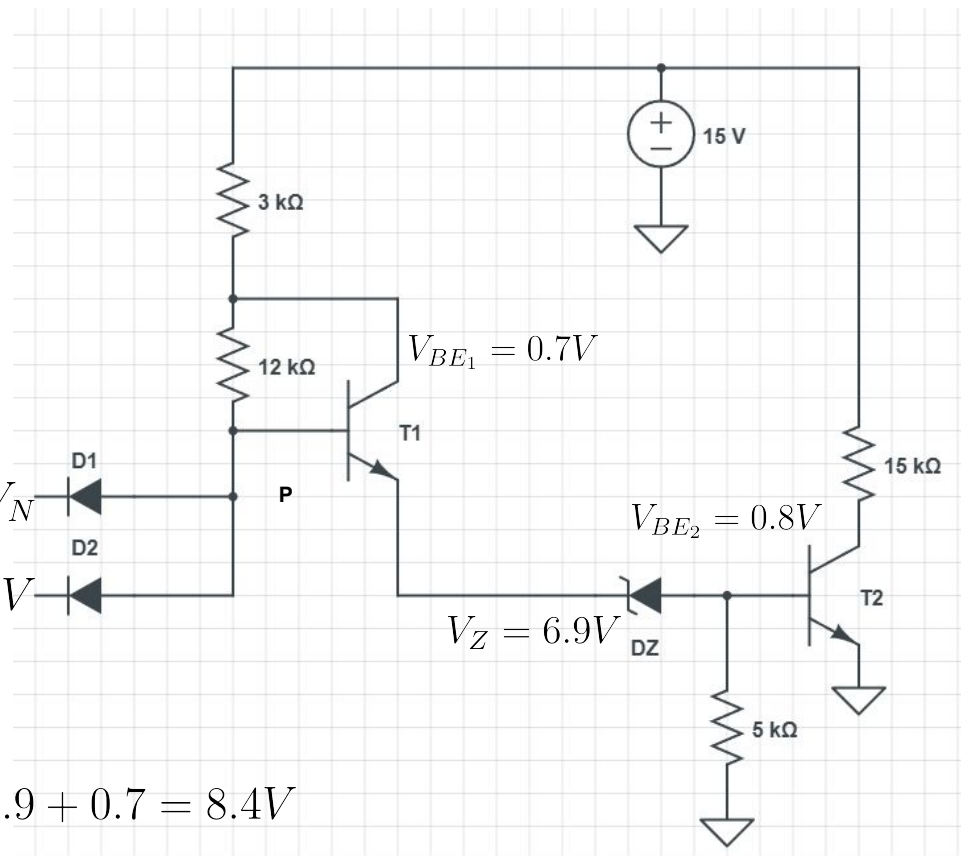
T2 will be in saturation & T1 in F.A.

Since, the voltage at cathode of the zener diode is higher, it will be reverse biased & have 6.9V across it. The base to emitter voltage of the transistors are shown in the figure. So,

$$V_A + V_N = 15 + V_N$$

$$V_B = 15V$$

$$V_P = 0.8 + 6.9 + 0.7 = 8.4V$$



Following a similar analysis as in the DTL circuit, the HTL circuit will malfunction when D_1 turns on. For that we have,

$$V_P - V_{\gamma D_1} = V_N + V_A$$

$$\Rightarrow 8.4 - 0.6 = V_N + 15$$

$$\Rightarrow V_N = -7.2V$$

Thus, considering only the magnitude, high noise margin increases to $V_{NH} = 7.2V$

Ex-2: *If all inputs are high except one, then what is the magnitude of noise voltage at that low input terminal, that would cause malfunction to the HTL circuit shown above?*

The logic case here is (0,1), assuming that low input is applied to the 'A' terminal. Then the following table can be found:

Component/ Parameter	D1	D2	Dz	T1	T2	V_A	V_o
State	On	Off	Off	Cutoff	Cutoff	$0.2+V_N$	15V

The circuit will malfunction when T1, T2 & Dz turns on. To turn on a transistor, we know the cut in voltage is 0.5V. Also, when Dz is on, it will have a 6.9V difference across it like in the previous example. So, through a similar analysis, we get,

$$V_P = 0.5 + 6.9 + 0.5 = 7.9V$$

$$\Rightarrow V_P - (V_A + V_N) = 0.7$$

$$\Rightarrow V_N = V_P - V_A - 0.7 = 7.9 - 0.2 - 0.7 = 7V$$

So, new low noise margin, $V_{NL} = 7V$

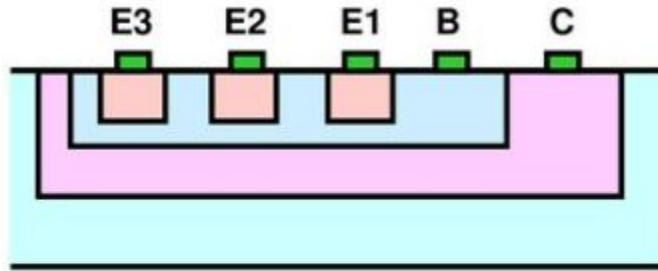
So, the overall noise margin of the HTL circuit is the minimum of the two margins = $\min(7.2, 7) = 7V \gg 0.8V$

Transistor Transistor Logic (TTL):

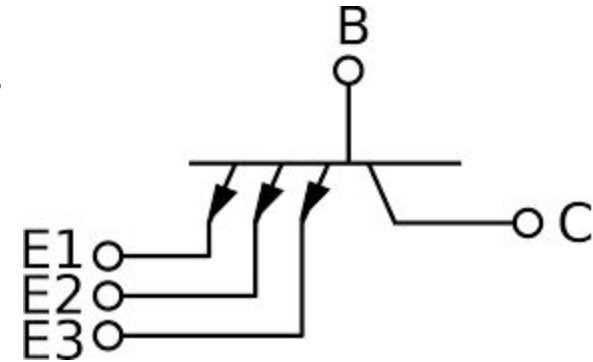
In this logic family, we use transistors at input instead of diodes. To discuss TTL circuits, first we need to know about Multi-Emitter Transistors.

Multi-Emitter Transistors: A BJT with multiple emitters is shown here.

The operating mode of the BJT depends on the emitter voltages. This transistor structure inside is something as shown below:

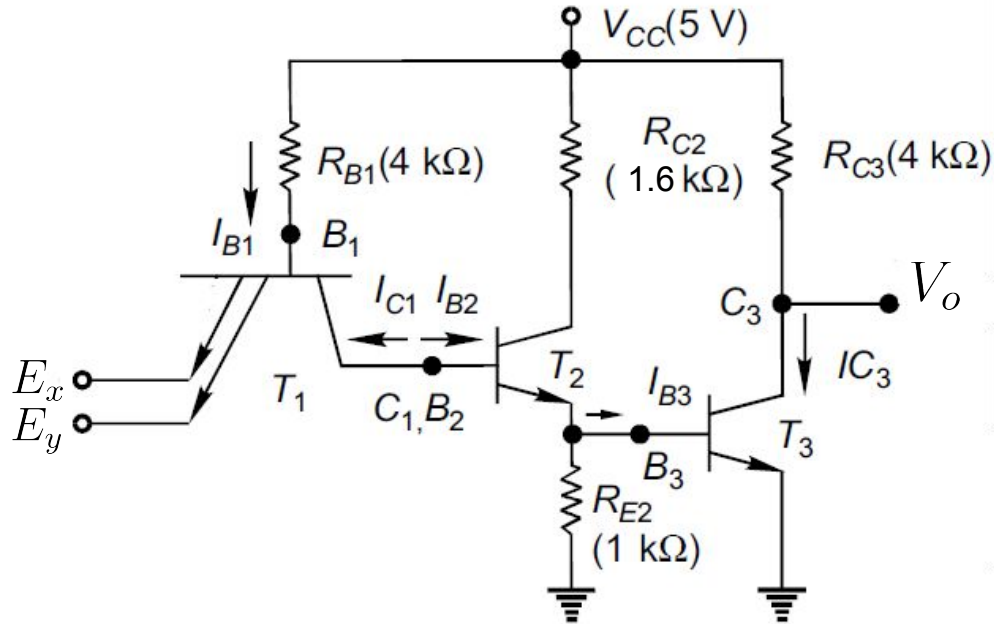


Physical Structure



The input emitter terminals are quite similar to the diode input terminals of DTLs, but not exactly the same.

Basic Operation of a TTL Circuit: Find out the currents & voltages involved in the following TTL circuit. Verify that it works like a NAND gate.



Case (0,0): Transistors T_2 & T_3 will be in cutoff, just like D_3 & D_4 in the DTL circuit (the two diodes at the base of the BJT in DTL). However, to understand what will be the state of T_1 , we need to analyze it a bit differently since it's not a regular BJT. The two emitter BJT can be visualized as follows:

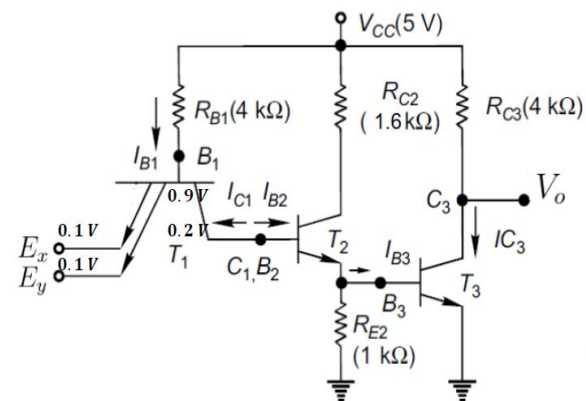
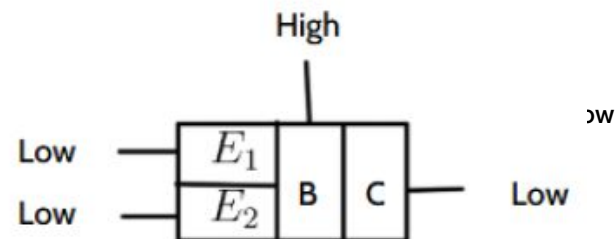
Thus, there are **two base-emitter junctions**. The voltage states in the four terminals are as shown in the figure. Thus, all the three junctions are forward biased. Thus, we can assume that the transistor is operating in **saturation** mode.

$$\text{So, } V_{CE_1} = 0.1 \text{ V}, V_{BE_1} = 0.8 \text{ V}$$

We see, the base voltage at T_2 is $0.2 \text{ V} < 0.5 \text{ V}$, which is not enough to turn on the transistor. Thus, our assumption is valid.

$$\text{Now, } I_{C_1} = -I_{B_2} = 0 \quad I_{C_2} = 0 = I_{E_2} = I_{C_3} = I_{E_3}$$

$$V_o = 5 \text{ V} \quad I_{B_1} = \frac{5 - 0.9}{4k} = 1.025 \text{ mA} \quad I_{E_{1x}} = I_{E_{1y}} = \frac{I_{B_1}}{2} = 0.5125 \text{ mA}$$



We find, $\beta_{forced} = \frac{I_{C1}}{I_{B1}} = \frac{0}{1.025} = 0 < \beta_F$

Thus, our assumption is valid.

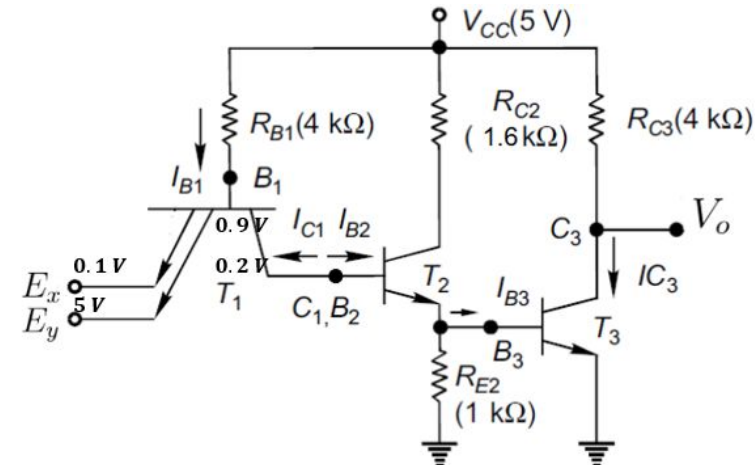
Case(0,1|1,0): This time, one of the emitter terminals of the multi-emitter transistor will have high voltage, and thus that B-E junction will be reverse biased, and off. Thus current from base will only flow through the low input emitter. However, since there is still one B-E junction left in forward bias, we can assume here too that the transistor is in saturation. The analyses are quite similar except for the emitter currents of the input transistor. We get:

$$I_{B1} = \frac{5 - 0.9}{4k} = 1.025mA$$

$$I_{E1x} = I_{B1} = 1.025mA$$

$$I_{E1y} = 0 \quad V_o = 5V$$

All other measurements are same as the previous case.



Case(1,1): When both inputs are high, both the B-E junctions will be reverse biased, while the B-C junction will be forward biased. This corresponds to Reverse Active mode. The emitters & the collector of input transistor will switch their roles.

The other two transistors can be assumed to be in saturation

(since output logic will be 0), and it can be proved.

In R.A., we know that, $V_{BC_1} = 0.7V$

Thus, $V_{B_1} = 2.3V$

$$I_{B_1} = \frac{5 - 2.3}{4k} = 0.675mA$$

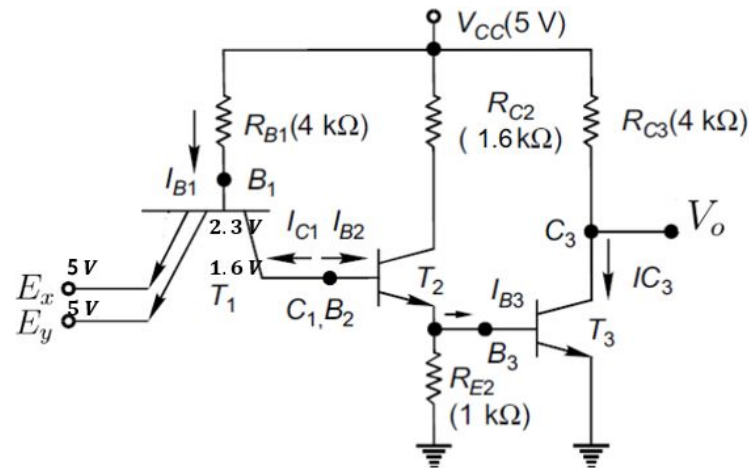
$$V_{CE_3} = 0.1V = V_{C_3}$$

$$V_{CE_2} = 0.1V$$

$$I_{C_3} = \frac{5 - 0.1}{4k} = 1.225mA$$

$$V_{C_2} = 0.8 + 0.1 = 0.9V$$

$$I_{C_2} = \frac{5 - 0.9}{1.6k} = 2.5625mA$$



Say it is given, $\beta_R = 0.1$

$$\beta_F = 25$$

Then,

$$I_{E_{1x}} = I_{E_{1y}} = I_{B_1} * \beta_R = 0.0675mA$$

$$I_{C_1} = I_{B_1} + I_{E_{1x}} + I_{E_{1y}} = 0.675 + 2 * 0.0675 = 0.81mA = I_{B_2}$$

$$I_{E_2} = I_{B_2} + I_{C_2} = 3.3725mA \quad I_{B_3} = I_{E_2} - \frac{0.8}{1k} = 2.5725mA$$

$$I_{E_3} = I_{B_3} + I_{C_3} = 3.7975mA$$

$$\beta_{forced_{T_3}} = \frac{I_{C_3}}{I_{B_3}} = 0.36 < 25$$

$$\beta_{froced_{T_2}} = \frac{I_{C_2}}{I_{B_2}} = 3.16 < 25$$

Thus our assumptions were valid.

Power Dissipation:

Case (0,0) & Case (0,1|1,0):

$$P = (5 - 0.1) \times I_{B_1} = 4.9 \times 1.025 = 5.0225 \text{ mW}$$

Case (1,1):

$$P = (5 - 0) \times [I_{B_1} + I_{C_2} + I_{C_3}] + (5 - 0) \times 2 \times I_{E_x} = 22.9875 \text{ mW}$$